

Predicting Ethylene Concentration Using ML

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Abstract—Accurate estimation of ethylene gas concentration is important in applications such as post-harvest storage, industrial safety, and environmental monitoring, where small deviations can affect product quality and human health. Conventional gas detection solutions based on specialized analyzers or single-purpose sensors are often expensive, difficult to scale, and sensitive to environmental disturbances, which limits their deployment in large-scale systems. This paper presents a machine-learning-based approach for predicting ethylene concentration using raw readings from a multivariate array of metal oxide semiconductor (MOS) and Taguchi gas sensors. The proposed pipeline performs systematic preprocessing including handling of missing values, outlier removal using the interquartile range (IQR) method, noise reduction via smoothing filters, and feature scaling, followed by feature engineering using correlation analysis and polynomial feature expansion. A polynomial regression model is then trained on the transformed features, and its performance is evaluated using K-fold cross-validation and regression metrics such as mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2). Experimental results on real sensor data show that the model can approximate the nonlinear relationship between sensor responses and ethylene concentration and deliver reliable predictions suitable for near real-time monitoring.

Index Terms—Ethylene gas, MOS sensors, Polynomial regression, Machine learning, Gas concentration prediction, Sensor fusion.

I. INTRODUCTION

Ethylene (C_2H_4) is a colorless and odorless gas that acts as a plant hormone and is widely encountered in industrial and environmental contexts. In post-harvest supply chains, ethylene concentration strongly influences fruit ripening and shelf life, while in industrial environments uncontrolled accumulation can lead to safety hazards and equipment damage. Continuous

monitoring of ethylene concentration is therefore essential for quality control, safety, and regulatory compliance.

Traditional ethylene measurement systems rely on specialized analyzers or highly selective sensors that provide accurate readings but are costly, bulky, and difficult to deploy at scale. In contrast, low-cost MOS and Taguchi gas sensors, such as TGS2610, TGS2620, TGS2600 and TGS2602, are readily available and easy to integrate with embedded platforms, but they suffer from nonlinearity, cross-sensitivity to other gases, temperature and humidity dependence, and drift. [1]–[4] These characteristics make direct mapping from raw sensor readings to ethylene concentration challenging.

Machine learning (ML) techniques enable sensor fusion and nonlinear regression, allowing the use of multivariate sensor arrays whose combined response encodes information about gas composition and concentration. By learning the mapping between sensor readings and target concentration, a data-driven model can compensate for limited selectivity and environmental effects. This work designs and implements a compact pipeline that predicts ethylene concentration using MOS sensor data with a focus on robust preprocessing, feature engineering, and a polynomial regression model offering a good trade-off between complexity, interpretability, and computational cost.

II. RELATED WORK

Low-cost gas sensing using MOS and Taguchi sensors has been explored for combustible gas detection, air-quality monitoring, and breath analysis, among other applications. These studies consistently report that individual MOS sensors exhibit nonlinear responses to gas concentration and are affected by ambient temperature, humidity, and cross-sensitivity

to multiple gases, motivating the use of sensor arrays and pattern-recognition techniques rather than simple thresholding.

Various machine learning models such as support vector regression (SVR), random forests, and artificial neural networks (ANNs) have been applied to estimate gas concentrations from sensor arrays, often achieving high accuracy but at the cost of increased computational complexity and reduced interpretability. For embedded and resource-constrained platforms, simpler models with fewer parameters are attractive as they reduce latency and ease deployment.

Polynomial regression extends linear regression by including higher-order terms of the input features, enabling the model to approximate nonlinear relationships while remaining relatively simple and interpretable. When combined with careful feature selection and preprocessing, polynomial regression can capture the nonlinear behavior of MOS sensor responses without incurring the full complexity of black-box models. The present work adopts such an approach and validates it using real ethylene sensor data.

III. SYSTEM OVERVIEW AND PROBLEM FORMULATION

The system considers an array of MOS/TGS sensors exposed to gas mixtures containing ethylene and other background gases. At each sampling instant t , a vector of raw sensor responses $\mathbf{S}(t) = [S_1(t), S_2(t), \dots, S_n(t)]$ is recorded together with the reference ethylene concentration $C_{C_2H_4}(t)$ obtained from calibrated instrumentation. The response of the i -th sensor can be modeled as

$$S_i(t) = f_i(C_{C_2H_4}(t)) + \varepsilon_i(t), \quad (1)$$

where $f_i(\cdot)$ is an unknown, generally nonlinear function specific to the sensor and $\varepsilon_i(t)$ captures noise and environmental variations such as temperature and humidity.

The objective is to learn a regression function

$$\hat{C}_{C_2H_4} = g(\mathbf{X}), \quad (2)$$

that predicts ethylene concentration from a feature vector $\mathbf{X}$ derived from the raw sensor readings, such that the error between $C_{C_2H_4}$ and $\hat{C}_{C_2H_4}$ is minimized under appropriate loss functions. Here $g(\cdot)$ is realized as a polynomial regression model fitted on transformed features produced by the preprocessing and feature-engineering pipeline.

Fig. 1 shows the high-level architecture from sensor array to model output.

IV. METHODOLOGY

The overall methodology follows a supervised ML pipeline consisting of data acquisition, preprocessing, feature engineering, model development, and evaluation. A block-level view of the processing stages is shown in Fig. 2.

A. Data Acquisition and Exploratory Analysis

The dataset contains multivariate sensor readings from an array of MOS and TGS sensors, along with ground-truth ethylene concentration values expressed in parts per million (ppm). Data were recorded under varying operating conditions

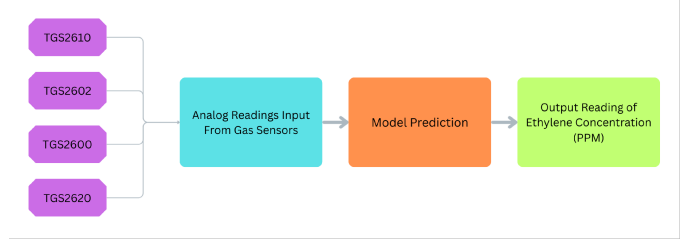


Fig. 1. High-level system architecture showing TGS gas sensors providing analog readings as input to the machine-learning model, which outputs the estimated ethylene concentration in ppm.

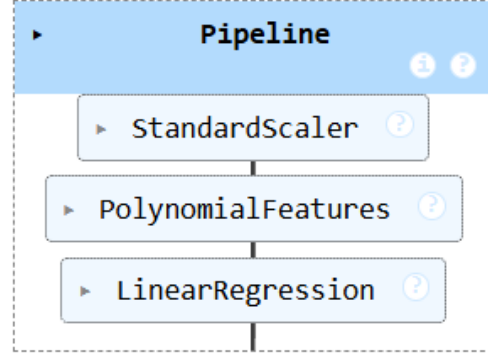


Fig. 2. Block diagram of the end-to-end pipeline, from raw data and exploratory data analysis to preprocessing, machine learning modelling and validation, and final model metrics and export.

so that the model would experience realistic variations arising from ambient temperature, humidity, and cross-sensitivity to other gases.

Exploratory data analysis (EDA) with histograms, density plots, and basic statistics was performed to understand the range, distribution, and variability of sensor readings and ethylene concentration. Correlation analysis using a heatmap provided insight into the relationships between sensor channels and ethylene concentration, revealing that some sensors are strongly correlated while others are weakly informative or redundant. A representative correlation heatmap is shown in Fig. 3.

B. Preprocessing

Raw sensor data contain missing values, outliers, and noise caused by sensor drift and environmental fluctuations.] To improve data quality, the following preprocessing steps were applied:

- **Handling missing values:** Records with missing entries were removed or imputed using suitable strategies to ensure consistent feature vectors.]
- **Outlier detection and removal:** The interquartile range (IQR) method was employed to detect extreme values which were then filtered out to prevent them from skewing the model.]

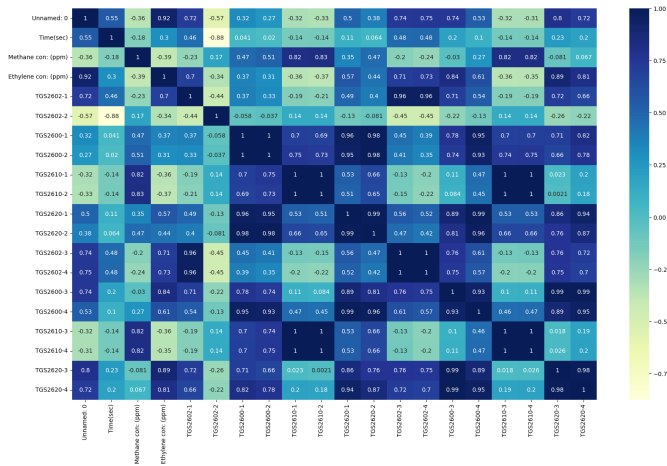


Fig. 3. Correlation heatmap of sensor features and target gas concentrations, highlighting sensors with strong positive or negative correlation to ethylene concentration and helping to guide feature selection.

- **Noise reduction:** A smoothing filter such as the Savitzky–Golay filter [5] was applied to reduce high-frequency noise in sensor readings.
- **Feature scaling:** Standardization was used to bring all features to a comparable scale, which improves numerical stability and ensures that features contribute proportionally to the regression model.]

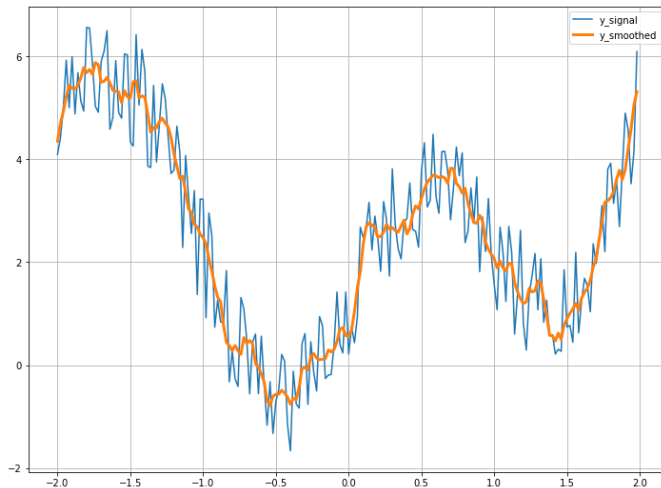


Fig. 4. Example of data smoothing applied to noisy sensor readings, showing the original signal and the smoothed signal after applying the chosen filter.

C. Feature Engineering and Polynomial Expansion

After preprocessing, feature engineering was used to construct a compact yet expressive representation of the data. Correlation analysis helped select the most informative sensor channels, and weakly correlated or highly redundant features were removed to reduce dimensionality and computation.

Polynomial feature expansion was applied to the selected features so that higher-order terms (e.g., squares and cross-products) were included as additional inputs.] This expansion enables a linear model in the expanded feature space to capture nonlinear relationships between sensor responses and ethylene concentration.

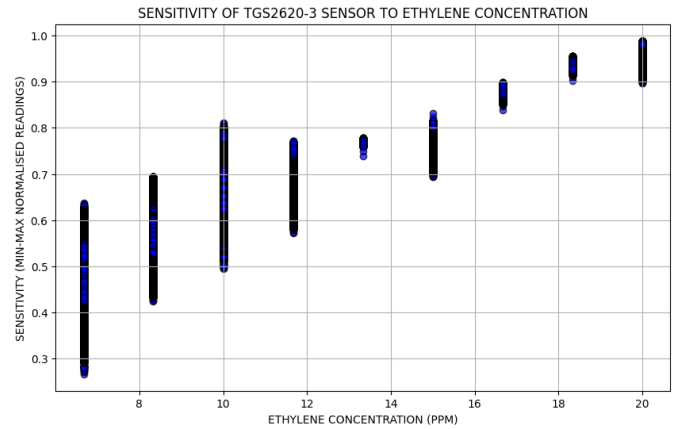


Fig. 5. Sensitivity of the TGS2620-3 sensor to ethylene concentration, with min-max normalised readings plotted against ethylene concentration (ppm), demonstrating an increasing nonlinear response.

D. Model Development and Validation

Polynomial regression was chosen as the core ML model because it can capture nonlinear relationships while remaining relatively simple, interpretable, and computationally efficient compared with SVR, random forests, or ANNs.] The model was implemented using `scikit-learn` [6], which provides utilities for polynomial feature generation, linear regression, and cross-validation.

The dataset was split into training and testing subsets, and polynomial degrees were varied to study the trade-off between underfitting and overfitting.] To obtain reliable performance estimates, K-fold cross-validation was employed: the data were partitioned into K folds, and in each run one fold served as validation data while the remaining folds were used for training. Regression metrics including MSE, RMSE, MAE and R^2 score were computed for each configuration and averaged over folds.]

V. IMPLEMENTATION AND EXPERIMENTAL RESULTS

A. Software Environment and Project Structure

The system was implemented in Python using widely adopted data-science libraries. Pandas was used for data loading and manipulation, NumPy for numerical computation,

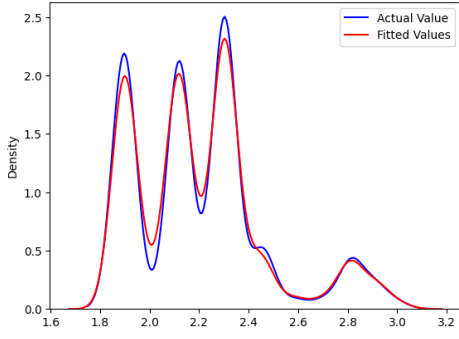


Fig. 6. Machine-learning pipeline used for ethylene prediction, consisting of standardization, polynomial feature expansion, and linear regression to realize the polynomial regression model.

SciPy for signal processing, scikit-learn for machine-learning components, and Seaborn and Matplotlib for visualization. [5]–[8] Jupyter Notebook served as the development environment and the project structure was organized into modular notebooks as shown in Fig. 7. Educational resources such as the “Analyzing Data with Python” course were referenced during the development of the pipeline. [9]

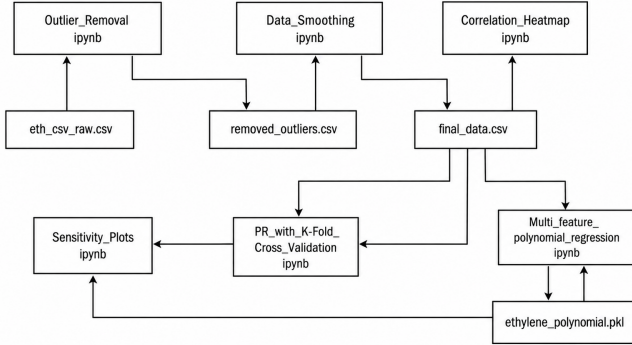


Fig. 7. Project structure illustrating the main notebooks and intermediate data files used for outlier removal, data smoothing, correlation analysis, training with K-fold cross-validation, generating sensitivity plots, and saving the final polynomial regression model.

A more detailed workflow emphasizing the interaction among modules is shown in Fig. 8.

B. Quantitative Performance

For the selected polynomial degree, the regression metrics on the processed dataset indicated low prediction error and high explained variance, demonstrating that the model generalizes reasonably well to unseen data. A distribution plot of actual versus predicted ethylene concentration shows that the predicted values closely follow the true distribution, with only small deviations at certain ranges of concentration; corresponding plots also show that the error distribution is narrow.]

The numerical summary of K-fold cross-validation, including average MSE, its standard deviation, average R-squared,

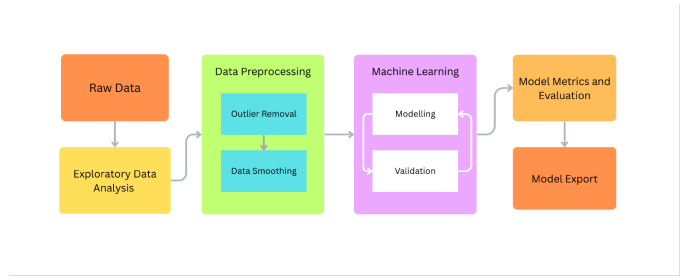


Fig. 8. Detailed workflow of the implemented system, indicating how preprocessing, modelling, validation, and evaluation stages interact to form an end-to-end solution.

and the corresponding error across folds, is summarised in Fig. 9.

Results :
Average Mean Square Error / Validation Loss : -0.001609589492804099
Std. Dev. of MSE : 0.00018086137292490674
Average R-Squared : 0.978033513878405
Average R-Squared Error : 0.02196648612159502

Fig. 9. Summary of K-fold cross-validation metrics for the polynomial regression model, including average mean squared error, its standard deviation, average R-squared, and R-squared error across folds.

C. Discussion

The experiments highlight the importance of a carefully designed preprocessing and feature-engineering pipeline when working with low-cost gas sensors. Outlier removal, noise reduction, and feature scaling significantly improved predictive performance compared with models trained directly on raw sensor data. Polynomial regression, combined with polynomial feature expansion, provided a good balance between prediction accuracy, interpretability, and computational efficiency, making it suitable for embedded or edge deployments.]

VI. CONCLUSION AND FUTURE WORK

This paper presented a compact machine-learning-based system for predicting ethylene gas concentration using multivariate MOS sensor data and polynomial regression. The pipeline covered EDA, preprocessing, feature engineering, model development, and evaluation, and demonstrated that a well-designed polynomial model can effectively capture the nonlinear behavior of sensor responses. The approach offers acceptable accuracy with relatively low computational requirements, which is attractive for real-time monitoring applications.

Future work includes extending the framework to multi-gas prediction, explicitly incorporating environmental variables into the model, and exploring more advanced yet efficient regression techniques such as ensemble methods or lightweight neural networks for additional performance gains. Integration with IoT platforms for remote monitoring and deployment on embedded hardware are also promising directions.

ACKNOWLEDGMENT

The authors would like to thank the Department of CSM(AI ML IoT), VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad, for providing facilities and guidance throughout this project.

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